

# 维修手册\_TC820LTYsi\_安全规程数据

维修手册知识库拆分数据 · 全字段 PDF 导出

|        |                                                             |
|--------|-------------------------------------------------------------|
| 源数据文件  | 维修手册_TC820LTYsi_安全规程数据.jsonl                                |
| 源文件路径  | D:/QQ\维修手册_TC820LTYsi_安全规程数据.jsonl                          |
| 记录条数   | 3 条                                                         |
| 字段并集   | 32 个字段（每条按分支实际字段渲染）                                         |
| 覆盖手册页码 | 78 – 84                                                     |
| 文档来源   | Southwestern Industries · TRAK TCsi with SINUMERIK ONE 维修手册 |
| 导出时间   | 2026-09-15 14:07:50                                         |

## 记录类别分布

| 记录类别          | 条数 | 占比     |
|---------------|----|--------|
| 安全规程 (safety) | 3  | 100.0% |

## 知识类型分布

| knowledge_type   | 条数 |
|------------------|----|
| safety_procedure | 3  |

说明：本 PDF 为该 JSONL 文件的逐条全字段渲染，未经任何筛选或删减；字段排序遵循 “标识与来源 / 型号与位置 / 分类与治理 / 业务明细 / 检索与原文” 的分组。空值以 “—” 表示，布尔值以 是/否 表示，数组按逐项换行展示。

| #1 lty-source-tcsi-safety-cleaning-pendant |                                       |      | 安全规程                                           |
|--------------------------------------------|---------------------------------------|------|------------------------------------------------|
| 部件                                         | touchscreen and machine control panel | 章节   | Maintenance / Cleaning the Touchscreen and MCP |
| 起始页                                        | 84                                    | 安全等级 | high                                           |
| 知识类型                                       | safety_procedure                      | 适用机型 | TC820LTYsi                                     |

**摘要 | 清洁操作面板前必须断电；只能用蘸水或温和清洁剂的软布清洁正面，不得直接喷液体，也不得使用压缩空气或蒸汽。**

A • 标识与来源

|      |                                                                                                  |
|------|--------------------------------------------------------------------------------------------------|
| 记录ID | lty-source-tcsi-safety-cleaning-pendant                                                          |
| 文档ID | trak-tcsi-service-34242-v022725                                                                  |
| 文档标题 | TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List |
| 来源类型 | maintenance_manual                                                                               |
| 源文件  | 维修手册/维修手册.pdf                                                                                    |
| 制造商  | Southwestern Industries, Inc.                                                                    |
| 文档版本 | 022725                                                                                           |
| 语言   | en                                                                                               |
| 产品系列 | TRAK TCsi                                                                                        |

B • 型号与位置

|      |                                                |
|------|------------------------------------------------|
| 设备型号 | TC820LTYsi                                     |
| 覆盖型号 | · TC820si<br>· TC820LTYsi                      |
| 型号范围 | TCsi-common                                    |
| 型号专属 | 否                                              |
| 章节编号 | 5.1.1                                          |
| 章节   | Maintenance / Cleaning the Touchscreen and MCP |
| 起始页  | 84                                             |
| 结束页  | 84                                             |

C • 分类与治理

|       |                                       |
|-------|---------------------------------------|
| 记录类别  | safety                                |
| 知识类型  | safety_procedure                      |
| 部件    | touchscreen and machine control panel |
| 安全等级  | high                                  |
| 可入库   | 是                                     |
| 需人工复核 | 否                                     |

D • 业务明细

|      |                                                   |
|------|---------------------------------------------------|
| 检查项  | · 确认设备已断电<br>· 确认清洁剂喷在布上而非面板上                     |
| 处置措施 | 用软布和水或温和清洁剂擦拭正面，用干布清洁背面；禁止腐蚀性溶剂、研磨性清洁剂、压缩空气和蒸汽喷射。 |

|           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 适用机型      | TC820LTYsi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| E • 检索与原文 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 中文摘要      | 清洁操作面板前必须断电；只能用蘸水或温和清洁剂的软布清洁正面，不得直接喷液体，也不得使用压缩空气或蒸汽。                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 关键词       | · 清洁<br>· 触摸屏<br>· MCP<br>· 断电<br>· 压缩空气<br>· 蒸汽                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 检索文本      | 清洁操作面板前必须断电；只能用蘸水或温和清洁剂的软布清洁正面，不得直接喷液体，也不得使用压缩空气或蒸汽。；；；用软布和水或温和清洁剂擦拭正面，用干布清洁背面；禁止腐蚀性溶剂、研磨性清洁剂、压缩空气和蒸汽喷射。；Turn power off when cleaning the pendant. Do not clean it using compressed air or steam jets. Clean only the front panel with a soft cloth moistened with water or mild cleaning agent; spray the agent or water on the cloth, never directly on the panel. Do not use caustic solvents or abrasive cleaners.；确认设备已断电；确认清洁剂喷在布上而非面板上；清洁；触摸屏；MCP；断电；压缩空气；蒸汽                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 原文摘录      | 5.0 Maintenance and Replacement 5.1 Maintenance 5.1.1 How to Clean the Touchscreen & Machine Control Panel (MCP) The pendant is designed for low maintenance operation. Nevertheless, you must regularly clean the pendant. WARNING!!! 1. Turn power off when cleaning the pendant. 2. Do not clean the device using compressed air or steam jets because they can damage it. Clean only the front panel with a soft cloth moistened with water of mild cleaning agent. Spray the cleaning agent or water on to a cleaning cloth. Do not spray anything directly on the front panel. Never use caustic solvents or abrasive cleaners. Use dry cloth to clean the backside of the pendant. 5.1.2 How to Clean the Electrical Cabinet Heat Exchanger Air Filter 1. Lift tab at the bottom of the filter which lifts the filter out of the slot in front of the fan, see figure 4.1.2. 2. Clean filter either by blowing with air or rinsing with water. 3. Clean fins on the grate of the fan. 4. Dry filter if needed and replace into the slot and slide down to secure. Figure 4.1.2 Heat Exchanger Filter |

| #2 lty-source-tcsi-diagnostic-collision |                     |      | 安全规程                                          |
|-----------------------------------------|---------------------|------|-----------------------------------------------|
| 部件                                      | collision detection | 章节   | TRAK Diagnostic Screens / Collision Detection |
| 起始页                                     | 78                  | 安全等级 | high                                          |
| 知识类型                                    | safety_procedure    | 适用机型 | TC820LTYsi                                    |

摘要 | 碰撞检测消息出现后先检查损伤和失准，继续加工前应谨慎，清除消息须联系 TRAK。

|           |                                                                                                  |
|-----------|--------------------------------------------------------------------------------------------------|
| A • 标识与来源 |                                                                                                  |
| 记录ID      | lty-source-tcsi-diagnostic-collision                                                             |
| 文档ID      | trak-tcsi-service-34242-v022725                                                                  |
| 文档标题      | TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List |
| 来源类型      | maintenance_manual                                                                               |
| 源文件       | 维修手册/维修手册.pdf                                                                                    |
| 制造商       | Southwestern Industries, Inc.                                                                    |
| 文档版本      | 022725                                                                                           |
| 语言        | en                                                                                               |
| 产品系列      | TRAK TCsi                                                                                        |

B • 型号与位置

|      |                                               |
|------|-----------------------------------------------|
| 设备型号 | TC820LTYsi                                    |
| 覆盖型号 | · TC820si<br>· TC820LTYsi                     |
| 型号范围 | TCsi-common                                   |
| 型号专属 | 否                                             |
| 章节编号 | 4.5.3                                         |
| 章节   | TRAK Diagnostic Screens / Collision Detection |
| 起始页  | 78                                            |
| 结束页  | 78                                            |

C • 分类与治理

|       |                     |
|-------|---------------------|
| 记录类别  | safety              |
| 知识类型  | safety_procedure    |
| 部件    | collision detection |
| 安全等级  | high                |
| 可入库   | 是                   |
| 需人工复核 | 否                   |

D • 业务明细

|      |                                                |
|------|------------------------------------------------|
| 故障现象 | 发生碰撞，控制器显示碰撞检测消息。                              |
| 原因   | 检测到轴或机床碰撞；碰撞转矩值由 TRAK 出厂设定且用户不可修改。             |
| 检查项  | · 记录碰撞日期和时间<br>· 检查机床是否有损伤或失准                  |
| 处置措施 | 谨慎停止继续加工；清除碰撞消息前联系 TRAK Customer Service 检查机床。 |
| 适用机型 | TC820LTYsi                                     |

E • 检索与原文

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 中文摘要 | 碰撞检测消息出现后先检查损伤和失准，继续加工前应谨慎，清除消息须联系 TRAK。                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 关键词  | · 碰撞<br>· Collision Detection<br>· 失准<br>· 损伤                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 检索文本 | 碰撞检测消息出现后先检查损伤和失准，继续加工前应谨慎，清除消息须联系 TRAK。；发生碰撞，控制器显示碰撞检测消息。；检测到轴或机床碰撞；碰撞转矩值由 TRAK 出厂设定且用户不可修改。；谨慎停止继续加工；清除碰撞消息前联系 TRAK Customer Service 检查机床。；Collision torque values are set at the factory and cannot be changed by the user. If a collision occurs, check the machine for damage or misalignment. The message does not prevent the machine from running; proceed with caution. Call TRAK Customer Service to check the machine before clearing the message.；记录碰撞日期和时间；检查机床是否有损伤或失准；碰撞；Collision Detection；失准；损伤 |

原文摘录

The control will then display the I/O Diagnostic screen see figure 3.5.2b. The I/O Diagnostic screen will show the current state of the machine’ s inputs and outputs. Different inputs and outputs can be toggled to and monitored on the I/O Diagnostic screen. The I/O diagnostic screen can be used as a tool to troubleshoot different functions on the machine and help determine where any issue may be. When on the phone with TRAK customer service they may ask you to go to the I/O diagnostic screen to help troubleshoot any issues the machine is having. Figure 3.5.2b I/O Diagnostic 4.5.3 Collision Detection The Collision Detection screen is found by pressing the left VSK “Diag” button, from “Diag” page 1 press “TRAK Diag.” button on the right side of the HSK. From the “Run Times” press “Col. Detect.” on the right side of the HSK see figure 3.5.3a. Figure 3.5.3a Collision Detection The control will then display the Collision Detection screen see figure 3.5.3b. The collision torque values are set by TRAK at the factory and cannot be changed by the user. The collision detection will show the date and time when a collision occurs, and a message will appear at the top of the screen to notify the user of a collision. If a collision occurs, check for any damage or misalignment in the machine. The message will not prevent the machine from running so proceed with caution before continuing machining. To clear

| #3 lty-source-tcsi-diagnostic-door |                  |      | 安全规程                             |
|------------------------------------|------------------|------|----------------------------------|
| 部件                                 | door interlock   | 章节   | Door Guards Not Working Properly |
| 起始页                                | 82               | 安全等级 | critical                         |
| 知识类型                               | safety_procedure | 适用机型 | TC820LTYsi                       |

摘要 | 防护门不能锁定时检查开关、继电器和互锁对中；门锁异常时严禁运行机床。

A • 标识与来源

|      |                                                                                                  |
|------|--------------------------------------------------------------------------------------------------|
| 记录ID | lty-source-tcsi-diagnostic-door                                                                  |
| 文档ID | trak-tcsi-service-34242-v022725                                                                  |
| 文档标题 | TRAK TCsi with Siemens SINUMERIK ONE CNC Safety, Installation, Maintenance, Service & Parts List |
| 来源类型 | maintenance_manual                                                                               |
| 源文件  | 维修手册/维修手册.pdf                                                                                    |
| 制造商  | Southwestern Industries, Inc.                                                                    |
| 文档版本 | 022725                                                                                           |
| 语言   | en                                                                                               |
| 产品系列 | TRAK TCsi                                                                                        |

B • 型号与位置

|      |                                  |
|------|----------------------------------|
| 设备型号 | TC820LTYsi                       |
| 覆盖型号 | · TC820si<br>· TC820LTYsi        |
| 型号范围 | TCsi-common                      |
| 型号专属 | 否                                |
| 章节编号 | 4.10                             |
| 章节   | Door Guards Not Working Properly |
| 起始页  | 82                               |
| 结束页  | 82                               |

C • 分类与治理

|      |        |
|------|--------|
| 记录类别 | safety |
|------|--------|

|       |                  |
|-------|------------------|
| 知识类型  | safety_procedure |
| 部件    | door interlock   |
| 安全等级  | critical         |
| 可入库   | 是                |
| 需人工复核 | 否                |

D • 业务明细

|      |                                                                                                                |
|------|----------------------------------------------------------------------------------------------------------------|
| 故障现象 | 防护门不能正确锁定或门报警。                                                                                                 |
| 原因   | 门顶开关、门关闭/释放继电器连接异常，或门与互锁错位。                                                                                    |
| 检查项  | <div><div>· 检查门顶部开关</div><div>· 检查 door close relay 和 door release relay 连接</div><div>· 检查互锁钥匙是否对准</div></div> |
| 处置措施 | 门不能正确锁定时禁止操作机床。                                                                                                |
| 适用机型 | TC820LTYsi                                                                                                     |

E • 检索与原文

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 中文摘要 | 防护门不能锁定时检查开关、继电器和互锁对中；门锁异常时严禁运行机床。                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 关键词  | <div><div>· 防护门</div><div>· 互锁</div><div>· door guard</div><div>· 门报警</div><div>· 禁止运行</div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 检索文本 | 防护门不能锁定时检查开关、继电器和互锁对中；门锁异常时严禁运行机床。；防护门不能正确锁定或门报警。；门顶开关、门关闭/释放继电器连接异常，或门与互锁错位。；门不能正确锁定时禁止操作机床。；Check the switch on top of the door, the door close relay and door release relay connections, and whether the key aligns with the interlock. If the door does not lock properly, do not operate the machine.；检查门顶部开关；检查 door close relay 和 door release relay 连接；检查互锁钥匙是否对准；防护门；互锁；door guard；门报警；禁止运行                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 原文摘录 | <p>4.9 E-Stop Error (Machine Is Disabled) E stop turns power off to axis and spindle motor when the e stop is pressed a message is displayed on the control as shown in figure 3.9. To clear this message, hold down the reset button for 1 second. If this fail to clear the error, first confirm that no e stops are engaged. In the cabinet check connections to estop control relay. If these are off, then call TRAK Customer Service at (800)-367-3165. Figure 3.9 Emergency stop In the electrical cabinet check connections to E-Stop control relay. If fail to clear issue call TRAK Customer Service at (800)-367-3165. 4.10 Door Guards Not Working Properly Check the switch on top of the door. Check connections to door close relay and door release relay labeled in the electrical cabinet. If the key does not line up with the interlock, the door may have moved out of alignment. Figure 3.10 shows the alarm message when door is not secured. Figure 3.10 Door Error Alarm WARNING!!! If the door does not lock properly, do not operate machine. 4.11 Cable Connections The TCsi machines use several cables to communicate between systems. It is often the case that what appears to be the failure of an electrical component is actually attributable to a poor connection. 1. Turn off and unplug the system from the wall. WARNING! Do not plug and unplug connectors with the system power on. This may cause damage to the connector board and harm to the technician. 2. Visually inspect the connections for excessive debris, moisture, or obvious damage. 3. Carefully clean any chips away from the connectors. 4. One-by-one, take out each connector and then plug them back in. Do the same at the computer/display. 5. Make sure the screws are tightened on each of the connectors. 4.12 Chip Conveyor Troubleshooting The chip conveyor is designed to evacuate chips from the machine envelope after cutting. If not used and maintained correctly issues can occur. When the conveyor is noisy check for a buildup of chips and clear any obstructions. If noise persists check the conveyor belt for damage and call TRAK Customer Service at (800)-367-3165.</p> |